Sustainable Analytics

Option2: Dashboard

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Sustainable Cities & Equity Dashboard

An Integrated SDG-Based Analytical Framework

1. Background and Rationale

One of the processes that characterize the 21 st century is urbanisation. Although cities create economic growth, invention, and a sharing of cultures, they also urbanize poverty, inequality, and environmental pressure. The availability of proper housing, clean water, sanitation, as well as social services, has been highly unequal, especially to urban populations who are at risk. Such differences pose serious questions of whether cities are becoming more sustainable and inclusive or it is growth that continues to support inequalities.

The 2030 Agenda of Sustainable Development acknowledges cities as the core of sustainable development especially SDG 11 Sustainable Cities and Communities. Nevertheless, the urban sustainability cannot be evaluated separately. Urban outcomes are highly determined by poverty, inequality, and access to basic services.

This undertaking consequently designs a Sustainable Cities & Equity Dashboard that aligns numerous SDGs to analyze the way in which urban sustainability touches on social and economic equity. The dashboard will allow analysis of space and time, as well as cross-goal analysis to serve evidence-based knowledge on the inclusive city development.

2. Link for the Dashboard

https://rpsingh1234.shinyapps.io/SDG_Dashboard/

3. Selection of SDGs, Targets, and Indicators

3.1 Selection of SDGs

Four SDGs were selected based on their direct relevance to urban equity and sustainability:

- **SDG 1 – No Poverty**
  Addresses economic deprivation, which disproportionately affects urban informal settlements.
- **SDG 6 – Clean Water and Sanitation**
  Essential for public health and urban liveability, especially in rapidly growing cities.
- **SDG 10 – Reduced Inequalities**
  Captures income and social disparities that shape access to urban opportunities.

- **SDG 11 – Sustainable Cities and Communities**
  Directly focuses on housing, transport, urban planning, and environmental conditions.

These goals together provide a multi-dimensional view of sustainable cities, combining economic, social, service-related, and spatial dimensions.

3.2 Selection of Targets

Within each SDG, only targets directly relevant to cities and equity were considered. Examples include:

- **SDG 1**: Targets related to poverty incidence and vulnerability
- **SDG 6**: Targets focusing on access to drinking water and sanitation
- **SDG 10**: Targets addressing income inequality and social inclusion
- **SDG 11**: Targets related to housing, basic services, and urban environment

Targets that were highly specialised, policy-specific or poorly reported across countries were excluded to ensure analytical consistency.

3.3 Selection of Indicators

From the selected targets, indicators were chosen using the following criteria:

1. Conceptual relevance to urban equity
2. Availability across a large number of countries
3. Temporal coverage allowing trend analysis
4. Quantitative comparability across regions

Examples of indicators include:

- Proportion of population below the international poverty line
- Access to safely managed drinking water services
- Income inequality measures
- Urban housing and service-related indicators

Indicators with excessive missing data or very limited country coverage were excluded to maintain dashboard responsiveness and interpretability.

4. Data Source and Initial Structure

4.1 Data Source

All data were obtained from the UN SDG Global Database, which compiles official SDG indicators reported by national statistical offices and international agencies. This ensures data reliability,

comparability, and alignment with global SDG monitoring standards.

4.2 Raw Data Structure

Each SDG dataset was downloaded separately and contained:

- Country or region identifiers
- SDG goal, target, and indicator codes
- Indicator descriptions
- Year-wise values
- Extensive metadata fields

The raw structure varied slightly across SDGs, requiring harmonisation before integration.

5. Data Cleaning and Pre-Processing

Data cleaning was a critical step to ensure consistency across goals and indicators.

5.1 Removal of Irrelevant Columns/Rows

Columns unrelated to analysis such as the targets and indicators were removed. Also the columns with no values were removed. This reduced dataset size and improved processing efficiency.

5.2 Standardisation of Variable Names

Column names were standardised across all datasets to ensure seamless merging. Key variables retained include:

- SDG_Goal
- Target
- SeriesCode
- SeriesDescription
- GeoAreaName
- TimePeriod
- Value

5.3 Handling Missing Values

Missing values were retained as NA rather than imputed. This decision preserves transparency and avoids introducing artificial trends into the analysis. Visualisations clearly indicate where data is unavailable.

5.4 Filtering of Selected Indicators

Only previously selected indicators were retained. This ensured that the dashboard focuses strictly on equity-relevant urban metrics and avoids clutter.

6. Temporal Transformation: Creation of 5-Year Periods

Annual SDG data often contains short-term fluctuations and reporting gaps. To improve interpretability, years were aggregated into 5-year periods, such as:

- 2000–2004
- 2005–2009
- 2010–2014
- 2015–2019
- 2020–2024

For each country-indicator combination, values within a period were averaged. This approach:

- Smoothens short-term volatility
- Enables clearer long-term trend analysis
- Improves comparability across countries

The resulting variable Period_5yr is used throughout the dashboard.

7. Merging of Datasets

After cleaning and transformation, all SDG-specific datasets were merged into a single long-format dataset.

7.1 Long Format Justification

The long format allows:

- Dynamic filtering by SDG, indicator, country, and time
- Cross-SDG comparisons
- Efficient use in Shiny reactive expressions

Each row represents:

One indicator × one country × one 5-year period

7.2 Final Dataset Storage

The merged dataset was saved as an RDS file (sdg_long.rds), which:

- Ensures reproducibility
- Reduces file size
- Allows fast loading during dashboard deployment

8. Dashboard Architecture and Design

The dashboard was built using R Shiny, combining analytical rigor with interactivity.

8.1 User Inputs

- SDG selection
- Indicator selection
- Country selection (maximum 5)
- 5-year period selection

8.2 Dashboard Tabs

1. **Trend Analysis**
   Displays temporal changes in indicators across countries.
2. **Comparison**
   Compares country performance across selected periods.
3. **World Map**
   Visualises spatial distribution of

indicators using choropleth mapping.

4. **SDG Comparison**
   Explores relationships between indicators from different SDGs.
5. **Data Table**
   Provides exact numerical values for transparency and verification.

Hover tooltips provide contextual details such as country name, SDG, indicator, period, and value.

9. Tools and Technologies Used

- **RStudio** for all the data cleaning and merging
- **Shiny** for interactive web applications
- **dplyr & ggplot2** for data manipulation and plotting
- **plotly** for interactivity
- **sf & rnaturalearth** for spatial data
- **shinyapps.io** for deployment

10. Conclusion

Through this Sustainable Cities and Equity Dashboard, it is possible to illustrate how many SDGs can be combined in order to evaluate an inclusive urban development. Through the associations of poverty, inequality, delivery of service and sustainability of the city, the dashboard shifts beyond the analysis of single goals and indicates a structural interrelationship.

To achieve analytical strength and usability, the official UN SDG data is used, the cleaning operation is transparent, aggregation is performed after 5 years, and the visualisation is interactive. The dashboard is a useful exploratory device that can be used to determine whether cities are growing more equitable and sustainable over time or not.

11. Declaration of AI

Gemini was used for improving the Dashboard Aesthetics

ChatGPT & Claude were used for solving the error in R code

Chatgpt was also used for taking colur references and improving the storyline for the dashboard

Claude was used to refine the language of the report wherever necessary.